IV. Random Variables (GSE 518)

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Idea: It is easier to work with numbers than with sets/events. Hence, we use random variables.

1. Random Variables

Definition:

Let $(S, \mathcal{B}, P)$ be a probability space. A random variable (r.v.) is a function from S to $\mathbb{R}$.

A random variable $X : S \to \mathbb{R}$ induces a new probability space $(S', \mathcal{B}', P')$ where:

- $S' \subseteq \mathbb{R}$
- $\mathcal{B}'$ is the same as before, consisting of:
  - all subsets of $\mathbb{R}$ if S' is countable
  - all intervals of $\mathbb{R}$ if S' is uncountable
- P' is based on the original probability function P where:

- If S' is countable:

$$P_X(X = x) = P(\{s \in S : X(s) = x\})$$

- If S' is uncountable:

$$P_X(X \in A) = P(\{s \in S : X(s) \in A\})$$

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Example:

Flip two coins:

$$S = \{HH, HT, TH, TT\}$$

Let $X(s) = \#$ of heads

$$X(HH) = 2, \quad X(HT) = 1, \quad X(TH) = 1, \quad X(TT) = 0$$

Note: $X(s) : S \to \mathbb{R}$

Note:

$$S' = \{0, 1, 2\} \quad \text{(new sample space)}$$

$$\mathcal{B}' = \text{all subsets of } S' \quad (S' \text{ countable}), \text{ i.e.,}$$

$$\mathcal{B}' = \{0, 1, 2, 01, 02, 12, \emptyset, S'\}$$

A new probability function $P_X(X = x)$:

$$P_X(X = 0) = P(\{s \in S : X(s) = 0\}) = P(TT) = \frac{1}{4}$$

$$P_X(X = 1) = P(\{s \in S : X(s) = 1\}) = P(HT \cup TH) = \frac{1}{2}$$

$$P_X(X = 2) = P(\{s \in S : X(s) = 2\}) = P(HH) = \frac{1}{4}$$

Definition:

Two random variables X and Y are identically distributed $X \sim Y$, if

$$P(X \in A) = P(Y \in A) \text{ for every } A \in \mathcal{B}'$$

Note: $X = Y \Rightarrow X \sim Y$ but $X \sim Y \not\Rightarrow X = Y$

Example:

Flip two coins. Let $X = \#$ heads and $Y = \#$ tails:

$$P(X = 0) = P(Y = 0) = \frac{1}{4}$$

$$P(X = 1) = P(Y = 1) = \frac{1}{2}$$

$$P(X = 2) = P(Y = 2) = \frac{1}{4}$$

$$\Rightarrow X \sim Y \text{ but } X \neq Y$$